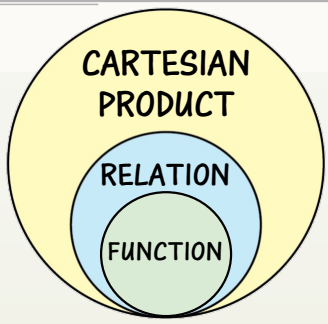
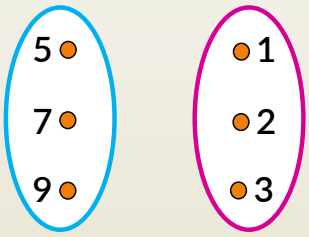
FUNCTIONS

Functions (Ch5.1)

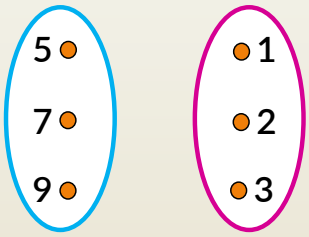
Function : a special type of relation



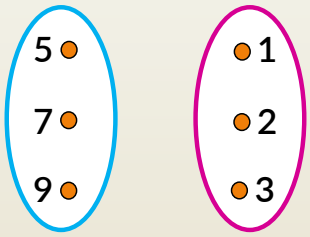
- A relation is a subset of the ***Cartesian*** product
- A ***relation*** can map inputs to ***multiple*** outputs
- A *relation* is a ***function*** when it maps each input to ***exactly one*** output



CARTESIAN PRODUCT
ALL PAIRS (UNIVERSAL REL.)

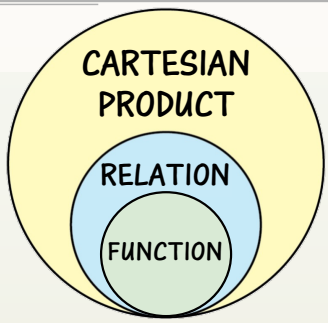


$3 < x - y < 6$



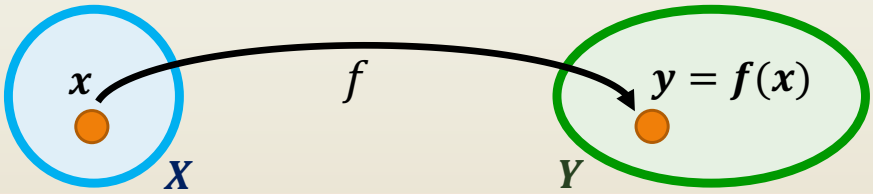
$y = x \bmod 4$

Function : a special type of relation



- A relation is a subset of the ***Cartesian*** product
- A ***relation*** can map inputs to ***multiple*** outputs
- A *relation* is a ***function*** when it maps each input to ***exactly one*** output

Let A and B be nonempty sets. A function f from A to B, denoted by $f: A \to B$, is an assignment of exactly one element of B to each element of A.

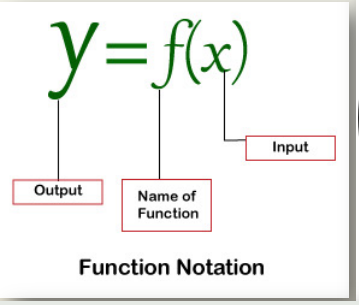


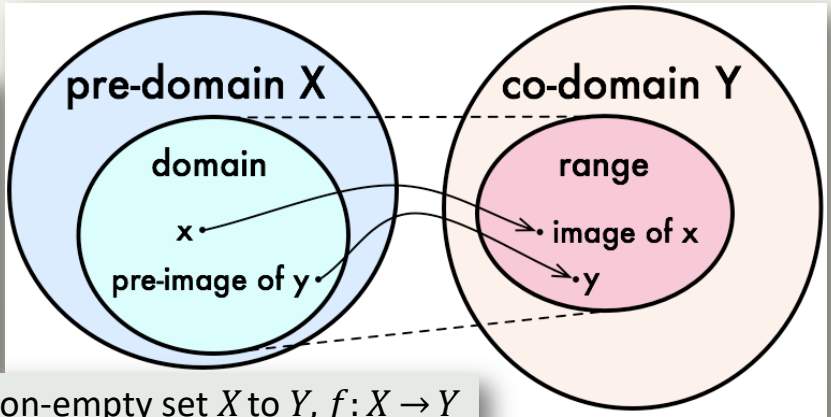
Domain, codomain, and range

Domain: the input; we call the output ***y the image of x under f***

Codomain: possible output

Range: actual output



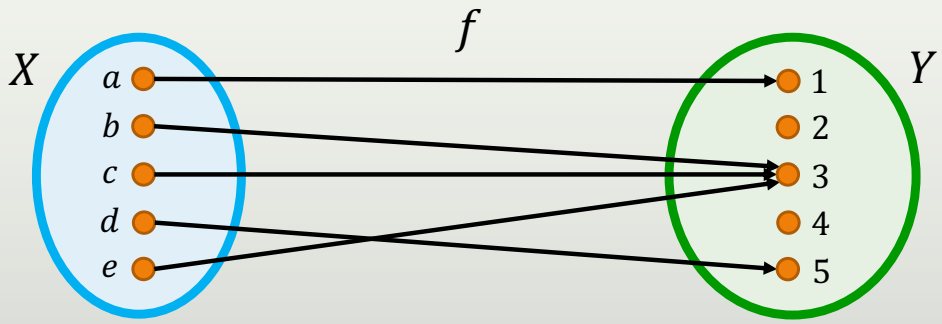


A function f from a non-empty set X to Y, $f: X \to Y$

https://ics.uci.edu/~alspaugh/cls/shr/correspondence.html

WORKED EXAMPLES

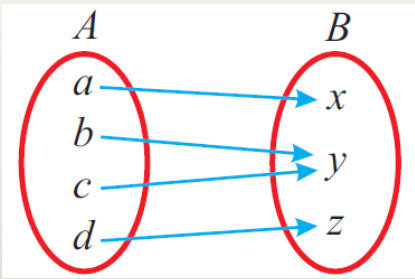
- Let $X = \{a, b, c, d, e\}$ and $Y = \{1,2,3,4,5\}$, f is a function from X to Y

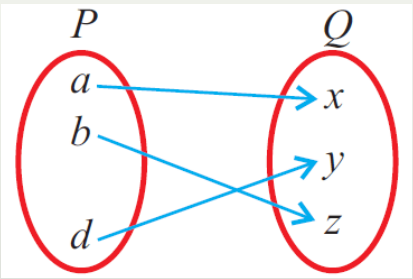


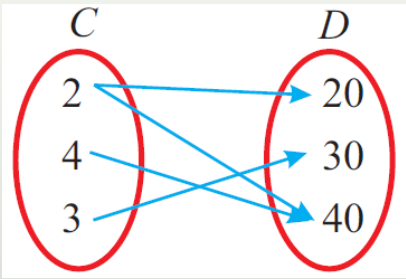
Domain (f)	Co-domain (f)	Range (f)	Image of e	Pre-image of 1

PRACTICE PROBLEMS

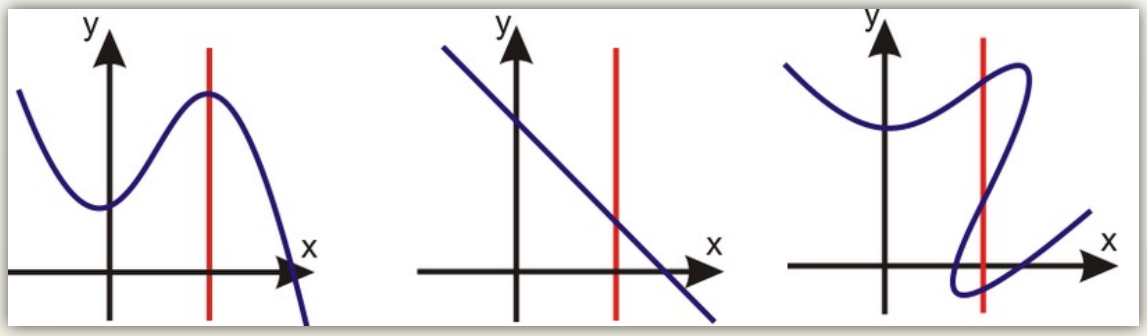
- Determine if the following relations represented as arrow diagrams are function. For one that is a function, give its domain, codomain, and range.







Vertical line test – is a relation from x to y a function?



The **vertical line test** states that a vertical line needs to cuts the graph of a function (equation) at only one point, for it to represent a function.

WORKED EXAMPLES

Given a relation $\boldsymbol{R}$ from $\mathbb{Z}$ to $\mathbb{Z}_{21}$, $\boldsymbol{R} = \{(\boldsymbol{x}, \boldsymbol{y}) \mid \boldsymbol{y} = \boldsymbol{x} \bmod 21\}$

Is R a function? If not, give a counterexample.

If so, what is the image of $\boldsymbol{x} = -56$ under $\boldsymbol{R}$?

Notation: $\mathbb{Z}_n$ denotes integer modulo n, $\mathbb{Z}_n = \{0,1,2, \ldots, n\text{-}1\}$

$\boldsymbol{xRy}$ iff $\boldsymbol{x}^2 + \boldsymbol{y}^2 = \mathbf{16}$ Give its domain and range.

Is R a function? If not, give a counterexample.

If so, what is the image of $\boldsymbol{x} = -3$ under $\boldsymbol{R}$?

PRACTICE PROBLEMS

For the given relation on $\mathbb{R}$: determine its domain and range. Then, decide whether it is a function. If your answer is 'no', give a counterexample showing why it is not.

- xRy iff $y \leq x - 1$
- xRy: $\begin{cases} y = 0 & \text{if } x \text{ is even} \\ y = 1 & \text{if } x \text{ is odd} \end{cases}$
- xRy iff $y = \frac{5}{x-1}$
  - Find image of 10
  - Find pre-image of 10

PRACTICE PROBLEMS

For the given relation on $\mathbb{R}$: determine its domain and range. Then, decide whether it is a function. If your answer is 'no', give a counterexample showing why it is not.

- xRy iff $y = \sqrt{x+5}$
- xRy iff $2x + 7y = 11$
- xRy iff $y = 2 - |x - 5|$
- $R = \{(x, y) \mid x \equiv y \bmod 21\}$

PRACTICE PROBLEMS

- Determine the domain, codomain, and range of $f: \mathbb{Z} \to \mathbb{Z},\ f(x) = x^2$

- For the given function, determine the image of the given x-value
  $f: \mathbb{Z} \to \mathbb{Z}$, $f(x) = \gcd(x, 24)$ and $x = 100$

PRACTICE PROBLEMS

- What will be the domain for which the functions $f(x) = 2x^2 - 1$ and $g(x) = 1 - 3x$ are equal?

- If $f(x) = ax + b$, where a and b are integers, $f(-1) = -5$ and $f(3) = 3$, then a and b are equal to

Combining functions using algebraic operations

Given f and g are functions

$$(f+g)(x) = f(x) + g(x)$$

$$(f-g)(x) = f(x) - g(x)$$

$$(fg)(x) = f(x)g(x)$$

$$\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}$$

WORKED EXAMPLES

- Let $A = B = \mathbb{Z}$, and C be the set of even integers. Let $f: A \to B$ and $g: B \to C$ be defined by $f(a) = a + 1$ and $g(b) = 2b$. Find
  - $(f+g)(x) =$
  - $(f-g)(x) =$
  - $(fg)(x) =$
  - $(f/g)(x) =$

WORKED EXAMPLES

- Let $A = B = \mathbb{Z}$, and C be the set of even integers. Let $f: A \to B$ and $g: B \to C$ be defined by $f(a) = a + 1$ and $g(b) = 2b$. Find
  - $(\boldsymbol{f} + \boldsymbol{g})(\mathbf{5}) =$
  - $(\boldsymbol{f} - \boldsymbol{g})(\mathbf{5}) =$
  - $(\boldsymbol{f}\boldsymbol{g})(\mathbf{5}) =$
  - $(\boldsymbol{f}/\boldsymbol{g})(\mathbf{5}) =$

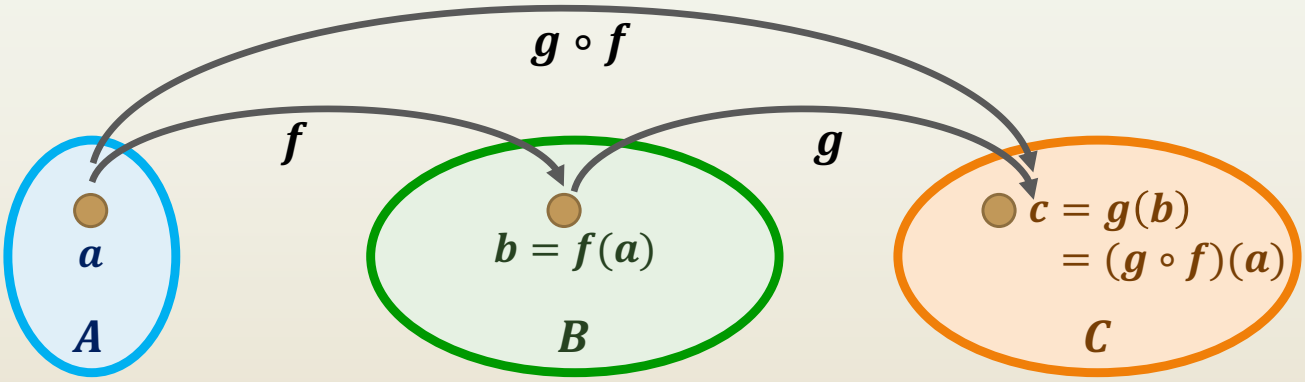
PRACTICE PROBLEMS

- If $f(x) = x^3 - (1/x^3)$, then $f(x) + f(1/x)$ is equal to

Composition of functions

$f: A \to B$ and $g: B \to C$ are functions.

The composition of f and g is a function $g \circ f = g(f(x))$

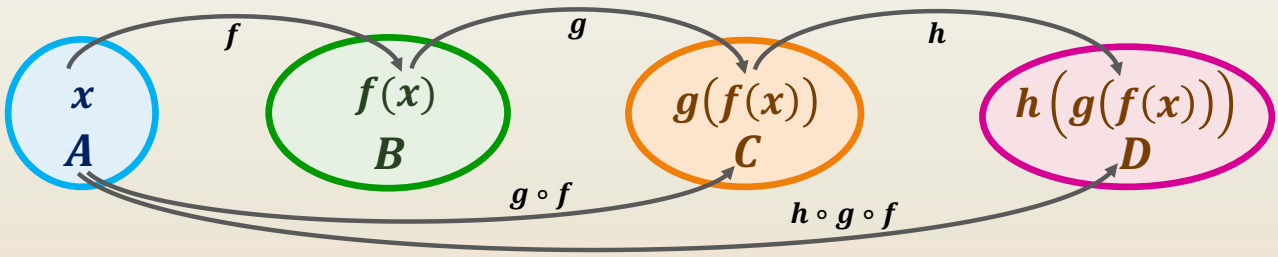


Composition of functions

$f(x) = 3x - 1$ and $g(x) = x^2$ and $h(x) = \frac{1}{x}$

$g \circ f =$

$h \circ g \circ f =$



WORKED EXAMPLES

Let f, g, and h be functions on $\mathbb{Q}$ (a set of rational numbers).
Let $f(x) = 3x - 1$ and $g(x) = x^2$ and $h(x) = \frac{1}{x}$

$(\boldsymbol{f} \circ \boldsymbol{g})(\boldsymbol{x}) =$

$(\boldsymbol{f}\boldsymbol{g})(\boldsymbol{x}) + (\boldsymbol{f} \circ \boldsymbol{g})(\boldsymbol{x}) =$

$(\boldsymbol{g} \circ \boldsymbol{h} \circ \boldsymbol{f})(\boldsymbol{x}) =$

$(\boldsymbol{f} \circ \boldsymbol{g} \circ \boldsymbol{h})\left(\frac{1}{5}\right) =$

PRACTICE PROBLEMS

- Let $A = B = \mathbb{Z}$, and C be the set of even integers. Let $f: A \to B$ and $g: B \to C$ be defined by $f(a) = a + 1$ and $g(b) = 2b$. Find ...
  - $(\boldsymbol{f} \circ \boldsymbol{g})(\boldsymbol{x}) =$
  - $(\boldsymbol{g} \circ \boldsymbol{f})(\boldsymbol{x}) =$
  - $(\boldsymbol{f} \circ \boldsymbol{f})(\boldsymbol{x}) =$
  - $(\boldsymbol{g} \circ \boldsymbol{g})(\boldsymbol{x}) =$
  - $(\boldsymbol{f} \circ \boldsymbol{g} \circ \boldsymbol{g})(\boldsymbol{x}) =$
  - $(\boldsymbol{f} \circ \boldsymbol{g})(\mathbf{5}) =$
  - $(\boldsymbol{g} \circ \boldsymbol{f})(\mathbf{5}) =$
  - $(\boldsymbol{f} \circ \boldsymbol{f})(\mathbf{5}) =$
  - $(\boldsymbol{g} \circ \boldsymbol{g})(\mathbf{5}) =$
  - $(\boldsymbol{f} \circ \boldsymbol{g} \circ \boldsymbol{g})(\mathbf{5}) =$

PRACTICE PROBLEMS

Let $A = B = \mathbb{Z}$, and C be the set of even integers. Let $f: A \to B$ and $g: B \to C$ be defined by $f(a) = a + 1$ and $g(b) = 2b$. Let $h: A \to B$, $h(c) = c^3$. Find ...

- $(\boldsymbol{f} \circ \boldsymbol{g} \circ \boldsymbol{h})(\boldsymbol{x}) =$
- $(\boldsymbol{h} \circ \boldsymbol{g} \circ \boldsymbol{f})(\boldsymbol{x}) =$
- $(\boldsymbol{g} \circ \boldsymbol{h} \circ \boldsymbol{f})(\mathbf{3}) =$

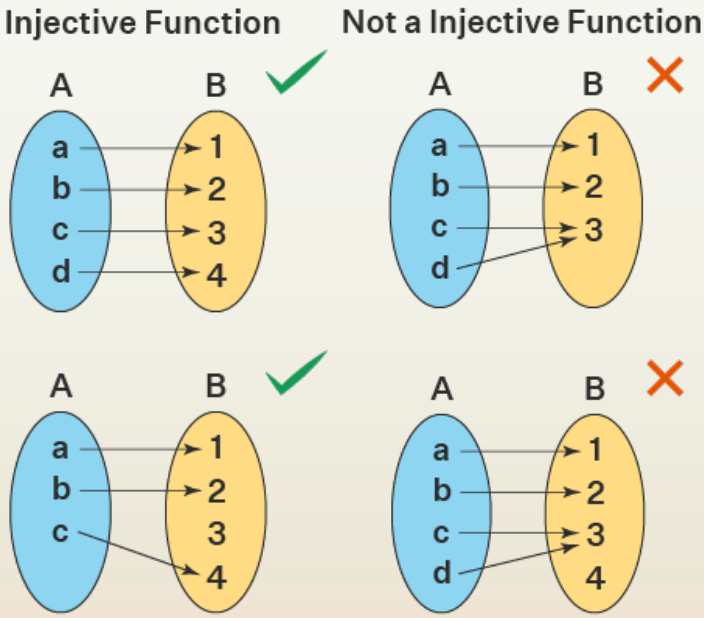
PRACTICE PROBLEMS

Let $f(x) = \frac{x}{x+3}$ and $g(x) = \frac{2}{x}$

- $(f+g)(x) =$
- $(f/g)(x) =$
- $(f \circ g)(x) =$
- $(g \circ f)(x) =$
- $(f \circ f)(x) =$
- $(g \circ g)(x) =$
- $(f+g)(5) =$
- $(f/g)(5) =$
- $(f \circ g)(5) =$
- $(g \circ f)(5) =$
- $(f \circ f)(5) =$
- $(g \circ g)(5) =$

One-to-one (injective) functions

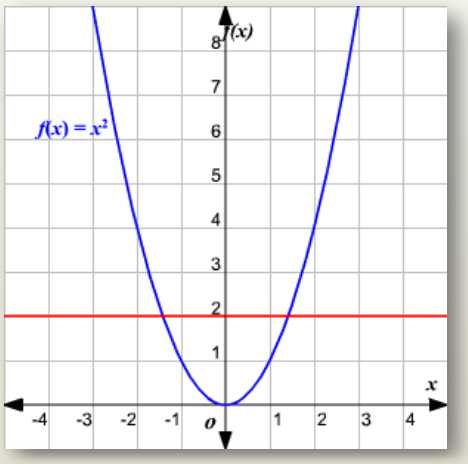
A function $f: A \to B$ is said to be **one-to-one** (or called **injective**), iff $f(a) = f(b)$ implies that $a = b$ for all a and b in the domain of f



https://www.cuemath.com/algebra/injective-function/

Checking one-to-one

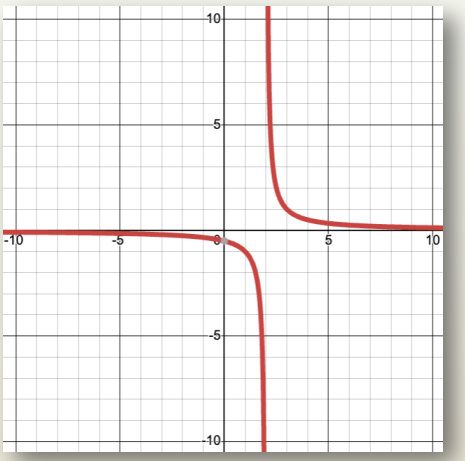
- Determine if $f: \mathbb{R} \to \mathbb{R}^+ \cup \{0\}$, $f(x) = x^2$ is one-to-one



*The **horizontal line test** states that if no horizontal line intersects the graph of a function f at more than one point, then f is one-to-one.*

Checking one-to-one

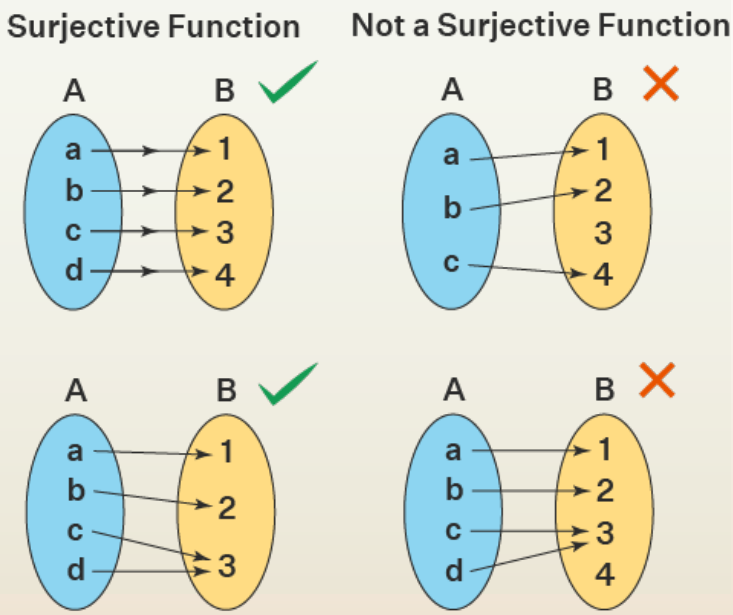
- Determine if $f: \mathbb{R} - \{2\} \to \mathbb{R}$, $f(x) = \frac{1}{x-2}$ is one-to-one



horizontal line test

Onto (surjective) functions

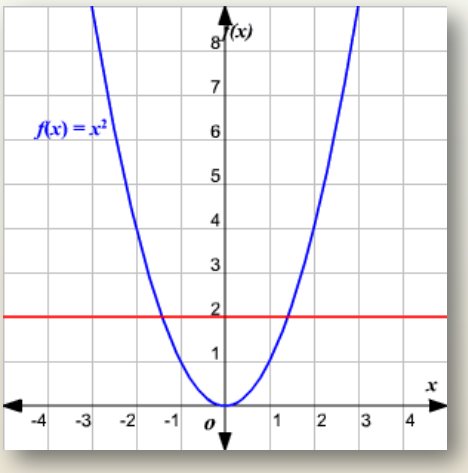
A function $f: A \to B$ is said to be **onto** (or **surjective**), iff for every element $b \in B$ there is an element $a \in A$ with $f(a) = b$



https://www.cuemath.com/algebra/injective-function/

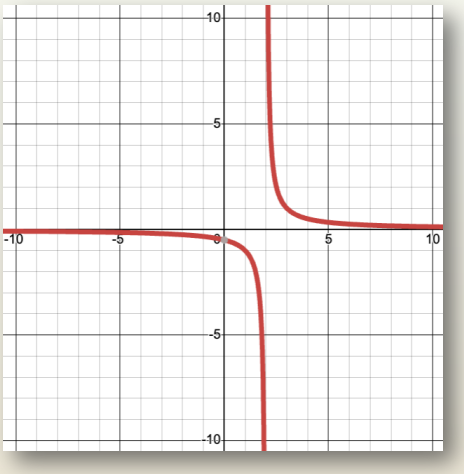
Checking onto

- Determine if $f: \mathbb{R} \to \mathbb{R}^+ \cup \{0\}$, $f(x) = x^2$ is onto



Checking onto

- Determine if $f: \mathbb{R} - \{2\} \to \mathbb{R}$, $f(x) = \frac{1}{x-2}$ is onto

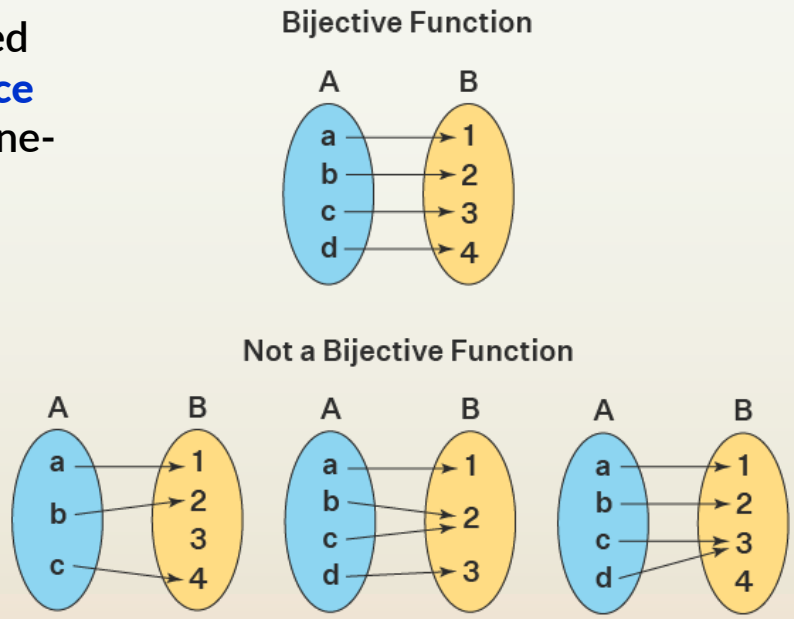


PRACTICE PROBLEMS

Determine if f on $\mathbb{R}-\left\{-\frac{1}{2},1\right\}$, $f(x)=\frac{2x}{1+2x}$ is one-to-one, onto, both, or neither

One-to-one correspondence (bijective) functions

A function $f: A \to B$ is called **one-to-one correspondence** (or **bijective**), if it is both one-to-one and onto



https://www.cuemath.com/algebra/bijective-function/

PRACTICE PROBLEMS

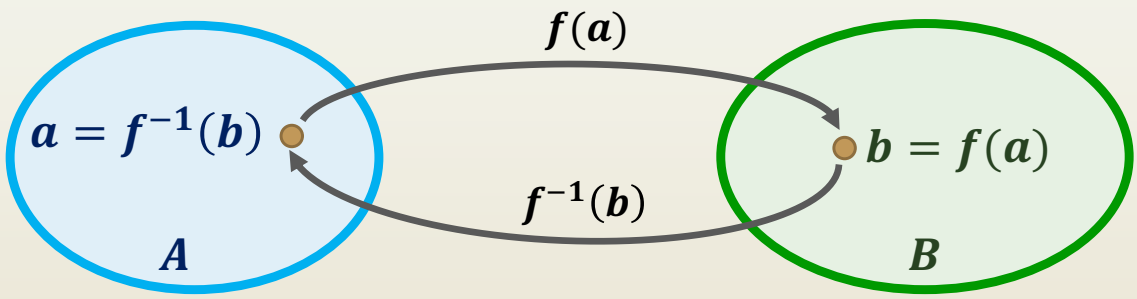
Choose only one option that best describes a given relation. Give a counter example for every property that a relation does not have.	not a func	bijective	one-to-one	onto	neither
$f: A \to B$, $A = \{1,2,3,4\}$, $B = \{a,b,c,d\}$, $f = \{(1,a),(2,b),(3,c),(3,d)\}$					
$f: \mathbb{Z} \to \mathbb{Z}_{100}$ and $f(x) = 5x \bmod 100$					
$f: \mathbb{R} \to \mathbb{R}^+$, $f(x) = x^2 - 2x + 1$					

PRACTICE PROBLEMS

Choose only one option that best describes a given relation. Give a counter example for every property that a relation does not have.	not a func	bijective	one-to-one	onto	neither
$f: \mathbb{R}\times\mathbb{R} \to \mathbb{R}$ and $f\big((x,y)\big) = x$					
$f: \mathbb{Z}^+ \to \mathbb{Z}^+$, $f(x) = \begin{cases} x+1 & \text{if } x \text{ is odd} \\ x-1 & \text{if } x \text{ is even} \end{cases}$					
$(g \circ f): \mathbb{N} \to \mathbb{N}$, $f(x) = x + 1$ and $g(x) = 2x + 3$					

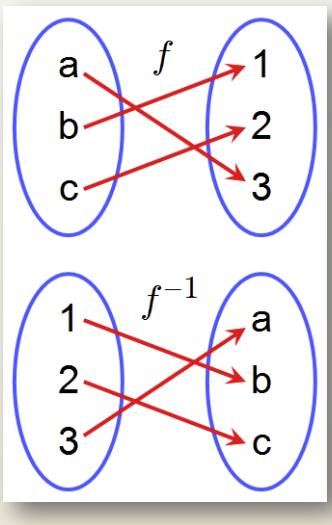
Invertible functions

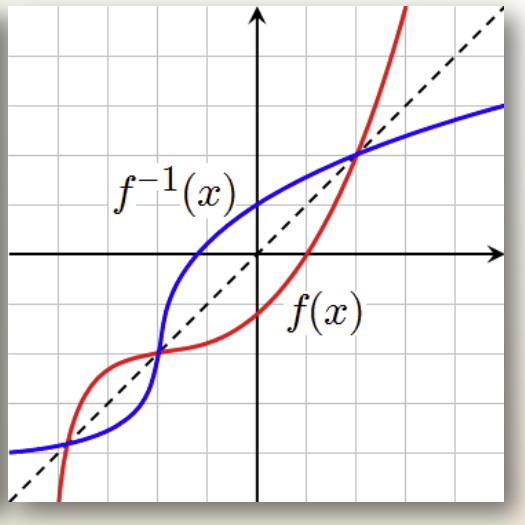
Let $f: A \to B$ be one-to-one correspondence. The inverse function of f assigns to an element b belonging to B the unique element a in A such that $f(a) = b$. The inverse function of f is denoted f^{-1}. Hence $f^{-1}(b) = a$ when $f(a) = b$.



Inverse of a function

- An inverse function reverses the inputs and outputs
- The graphs are reflections of each other over the line y=x



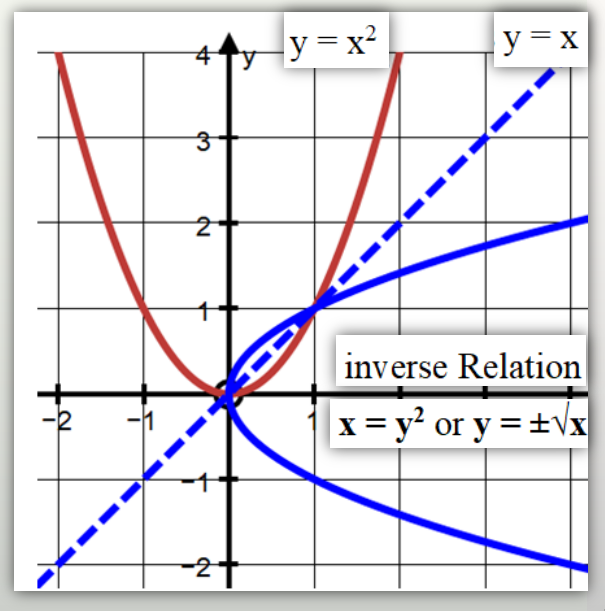


https://www.coursesidekick.com/mathematics/study-guides/boundless-algebra/inverse-and-composite-functions

WORKED EXAMPLES

Find the inverse of $f(x) = x^2$

Is f^{-1} a function?



WORKED EXAMPLES

Determine if $f: \mathbb{Z} \to \mathbb{Z}$, $f(x) = x + 1$ is invertible. If it is, find $f^{-1}(x)$

PRACTICE PROBLEMS

Find the inverse of f. Is $f^{-1}(x)$ a function?

$A = \{1,2,3,4\}$, $B = \{a, b, c, d\}$, and $f = \{(1, a), (2, a), (3, d), (4, c)\}$

PRACTICE PROBLEMS

Consider relations on $\mathbb{R}$. Let $f(x) = \frac{x-1}{x-3}$, $x \neq 3$ and $g(x) = \frac{x-3}{x-1}$, $x \neq 1$.

On a suitable domain, find $f^{-1}(x)$ and $g^{-1}(x)$.

PRACTICE PROBLEMS

Consider relations on $\mathbb{R}$. Let $f(x) = \frac{x-1}{x-3}$, $x \neq 3$ and $g(x) = \frac{x-3}{x-1}$, $x \neq 1$.

On a suitable domain, find $(f \circ g)(x)$ and $(g \circ f)(x)$.

PRACTICE PROBLEMS

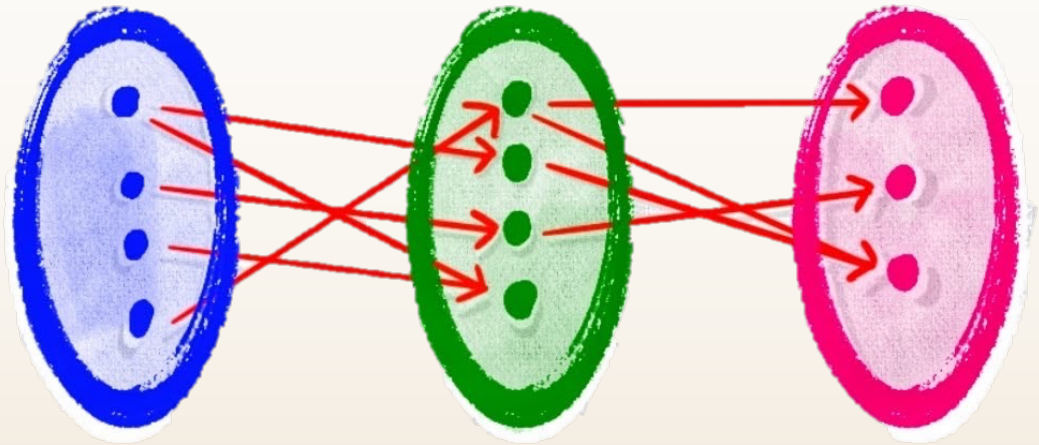
Consider relations on $\mathbb{R}$. Let $f(x) = \frac{3x+4}{x-2}$, $x \neq 2$. Find a function $g(x)$ on a suitable domain such that $(g \circ f)(x) = x = (f \circ g)(x)$.

PRACTICE PROBLEMS

- Let $f(x) = 10 - 2x$ and $g(x) = \frac{10-x}{2}$ and $h(x) = \sqrt{\frac{x}{2}}$
  - Find $f^{-1}(x)$
  - Find $g^{-1}(x)$
  - Find $f \circ g(x)$
  - Find $g \circ f(x)$
  - Find $h \circ f \circ g^{-1}(x)$

PRACTICE PROBLEMS

- Let $f(x) = 2x + 5$, find $f^{-1}(7)$
- Let $f(x) = \frac{x-5}{4}$, find $f^{-1}(1)$
- Let $f(x) = \frac{x-1}{2+3x}$, find $f^{-1}(x)$



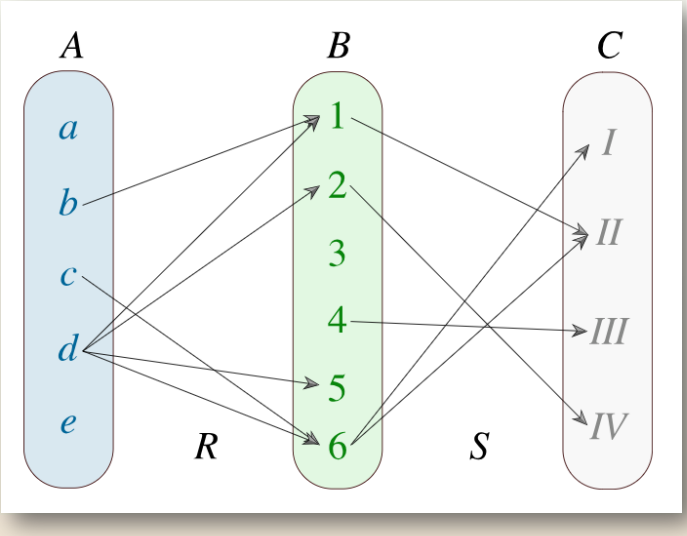
Composition of Relations

Composition of Relations

- Let A, B and C be three sets.
  R is a relation from A to B,
  S is a relation from B to C.
- The composition of R and S, denoted by S∘R, is

$S \circ R = \{(a,c) \mid \exists b \in B: aRb \wedge bSc\}$

= _____



WORKED EXAMPLES

Relations R = {(1,1),(1,3),(2,1),(2,2)} and S = {(1,2),(2,3),(3,1),(3,3)} on the set A = {1,2,3}. Perform the following operations:

- $R \cap S =$
- $R - S =$
- $R \circ S =$
- $S \circ R =$
- $R^{-1} \circ S =$

PRACTICE PROBLEMS

Given the following relations, perform each specified operation.
$R = \{(x,y) \mid xy > 0, x,y \in \mathbb{R}\}$ and $S = \{(x,y) \mid xy < 0, x,y \in \mathbb{R}\}$

- $R \circ S =$
- $S \circ R =$

Composition of relations in matrix form

If R and S are relations on a set A, then

$$\boldsymbol{M_{S\circ R} = M_R \odot M_S}$$

WORKED EXAMPLES

Find the matrix representation of the relation $\boldsymbol{S\circ R}$ when both $\boldsymbol{R}$ and $\boldsymbol{S}$ are defined on the set {1,2,3} and draw its digraph

$$M_R = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \text{ and } M_S = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

PRACTICE PROBLEMS

Find the matrix representation of the relation $\boldsymbol{R}^{-1} \circ \boldsymbol{S}$ when both $\boldsymbol{R}$ and $\boldsymbol{S}$ are defined on the set {1,2,3} and draw its digraph

$$M_R = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \text{ and } M_S = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

PRACTICE PROBLEMS

A = {a,b}, B = {0,1,2}, C = {x,y}. Relations R from A to B: R = {(a,0),(a,2),(b,1)}, S from B to C: S = {(0,x),(0,y),(1,y),(2,y)}. Find $M_{S \circ R}$ and draw its digraph.